Dallas International School

Test 3

Topics 2.2-2.3-2.4

11th IB - Math AA

For HL Students

Name : . . . . . .

EXERCISE 1 Write in simplest form :

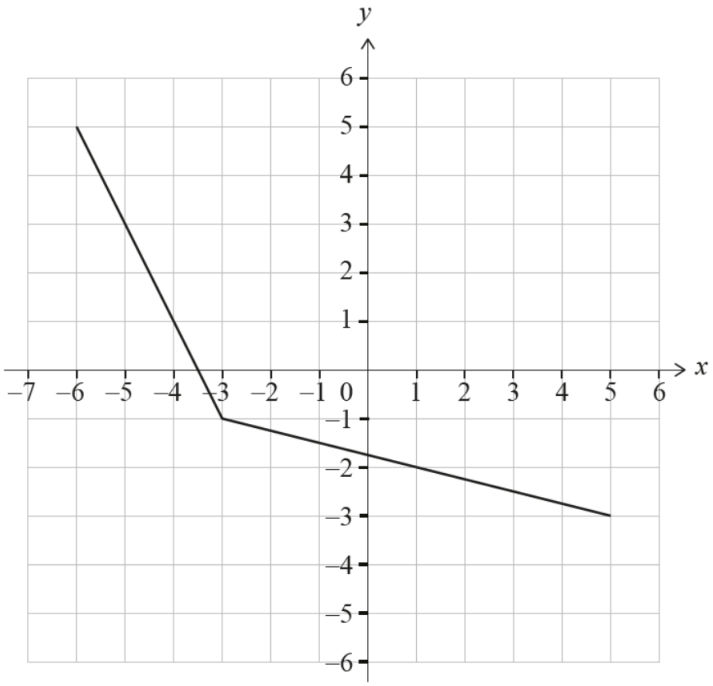
1. $\frac{6^n + 12^n}{1 + 2^n}$

2. $\frac{5^{n+1} - 5^n}{5^n}$

3. $(4^x + 7)^2$

4. $27^{-\frac{4}{3}}$

EXERCISE 2 The following diagram shows the graph of $y = f(x)$, for $-6 \leq x \leq 5$.



(a) Write down the value of $f(-3)$. [1]

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(b) State the domain of f^{-1}, the inverse function of f. [1]

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(c) Find the value of x that satisfies $f^{-1}(2x - 7) = -3$. [3]

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EXERCISE 3 A function f is defined by

$$f(x) = \frac{2x-1}{x+1}, \quad x \in \mathbb{R},\ x \neq -1.$$

The graph of $y = f(x)$ has a vertical asymptote and a horizontal asymptote.

(a.i) Write down the equation of the vertical asymptote. [1]

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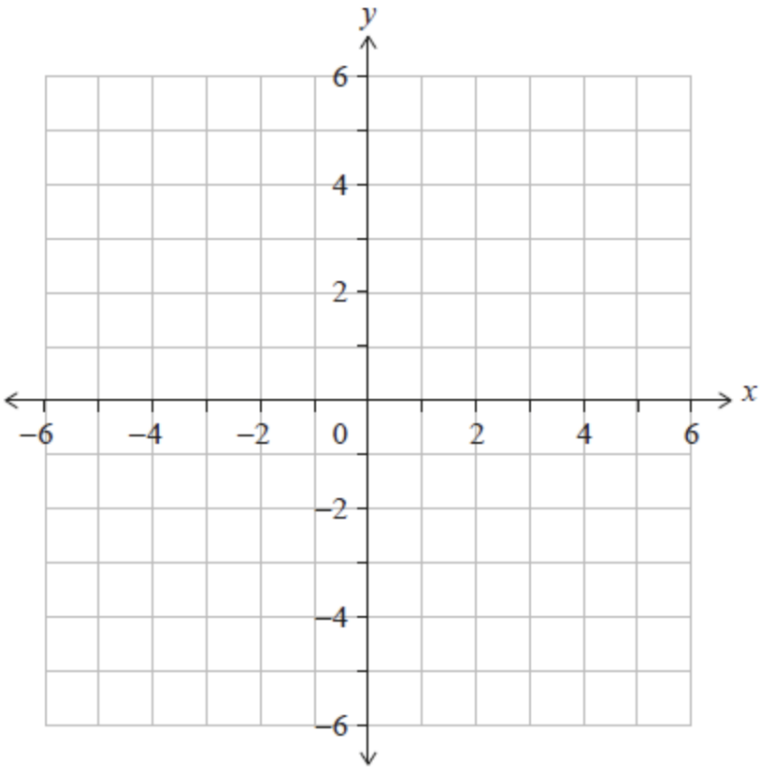
(a.ii) Write down the equation of the horizontal asymptote. [1]

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(b) On the set of axes below, sketch the graph of $y = f(x)$.
On your sketch, clearly indicate the asymptotes and the position of any points of intersection with the axes. [4]



(c) Hence, solve the inequality :

$$0 < \frac{2x-1}{x+1} < 2.$$

[1]

(d) **BONUS** Solve the inequality :

$$0 < \frac{2|x|-1}{|x|+1} < 2.$$

[2]

EXERCISE 4 A function g is defined by

$$g(x) = \frac{1}{x^2 - 2x - 3}, \quad x \in \mathbb{R},\ x > 3.$$

The inverse of g is g^{-1}.

1. Show that

$$g^{-1}(x) = 1 + \frac{\sqrt{4x^2 + x}}{x}.$$

[6]

2. State the domain of g^{-1}. [1]

EXERCISE 5 Consider the functions $f(x) = x + 2$ and $g(x) = x^2 - k^2$, where k is a real constant.

(a) Write down an expression for $(g \circ f)(x)$ [2]

(b) Given that $(g \circ f)(4) = 11$, find the possible values of k. [3]